



Technical Service Bulletin

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K Series Cylinder Head Changes (T206)

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Note : This is an update of Service Parts Topic 94T2-3. Remove and replace Service Parts Topic (SPT) 94T2-3 with this new topic. This topic releases the following new parts for K Series cylinder heads.

- Exhaust Valves, Part No. 3088389 and 3088391, and Intake Valve Part No. 3088393 have chrome plated valve stems for increased durability. In addition, all new K valves will also be chrome plated. See Table VIII, Part No. 3088389 replaces Part No. 3803528, Part No. 3088391 replaces Part No. 3803529, and Part No. 3088393 replaces Part No. 3803515.
- Cylinder head assemblies are now produced **without** valve crosshead guide. The valve crosshead guide is **not** required with the release of the stemless valve crosshead, Part No. 3086362. The valve crosshead guide hole is still machined in the cylinder head casting, for serving heads with stemmed crossheads, but the stemless crosshead can be used on all K Series engines.
- Valve Spring, Part No. 3630628 has been redesigned to stop wear at the first active coil that can progress to a fatigue failure. Part No. 3630628 replaces Part No. 3629007.

Premium No. 1: Cylinder head assembly, Part No. 3811985 is for use on high load factor and constant speed applications such as marine, base load or prime power generator drives, some drilling machines, and hydraulic excavators. This cylinder head assembly contains powdered metal valve guides, valve rotators, shorter springs, wear resistant Tribaloy intake valves and valve inserts, and high temperature Inconel exhaust valves. Reference Tables 1, Premium No. 1, for further part number information.

Premium No. 2: Cylinder head assembly, Part No. 3811986 is for use on uprate engines that do **not** require the Tribaloy intake valve. This cylinder head is used on all K Series engines with a maximum design rating of 101 horsepower (hp) per cylinder or more that do **not** require the Premium No. 1 cylinder head assembly, Part No. 3811985. The difference between the two cylinder head assemblies is Premium No. 2, Part No. 3811986 does **not** have the Tribaloy valves and valve inserts. Reference Table 1, Premium No. 2, for further part number information.

Improved Standard Cylinder Head: Improved standard cylinder head assembly, Part No. 3811987 is for use on all K Series engines with a maximum design rating of 100 hp per cylinder or less but **not** prime power or base load units. This assembly contains standard intake valves, Stellite 6 exhaust valves, and standard valve seat inserts but does **not** contain rotators. Reference Table 1, Standard, for further part number information.

Standard Cylinder Head: Cylinder head assembly, Part No. 3811988 is used in standby generator engines **not** prime power or base load units. This assembly contains standard intake valves, Stellite 6 exhaust valves, and standard valve seat inserts but does **not** contain rotators. Reference Table 1, Standard, for further part number information.

Special Application Cylinder Heads: Cylinder head assembly, Part No. 3811989 is for use on locomotive applications and was used on some hydraulic excavators. It contains standard valves, flat-top cast iron valve guides **with seals on all four valve guides**, but does **not** contain rotators. These valve seals are used in applications where there is a large percentage of idle time that can cause oil or fuel slobber. Reference Table 1, Special Application, Part No. 3811989, for further part number information.

Cylinder head, Part No. 3630361 is for use on horizontal rail car applications only. It contains specially machined valve guides for improved valve stem lubrication on horizontally mounted K19 engines. This part number replaces Part No. 3811990 that did **not** include a valve stem seal. Reference Table 1, Special Applications, Part No. 3630361 for further part number information.

Note : When installing a valve guide, Part No. 3235286, the oil hole in the guide **must** be pointed toward the closest crosshead guide or crosshead guide bore so the valve guide and stem will receive adequate lubrication.

TABLE 1 - Cylinder Head Part Number Information

Description	Premium No. 1	Premium No. 2	Improved Standard	Standard	Special Applications	
					3811989	3630361
Head Assembly	3811985	3811986	3811987	3811988	3811989	3630361
Head Casting	3081064	3081064	3081064	3081064	3081064	3081064
Valve Guide	3202210	3202210	3202210	3202210	3018818	3235286
Intake Valve	3052820	3088393	3088393	3088393	3088393	3088393
Intake Seat	3052819	3026296	3026296	3026296	3026296	3026296
Exhaust Valve	3088389	3088389	3088391	3088391	3088391	3088389
Exhaust Seat	205093	203093	205093	205093	205093	205093
Valve Spring	3630628	3630628	3630628	3629006	3629006	3630628

Description	Premium No. 1	Premium No. 2	Improved Standard	Standard	Special Applications	
					3811989	3630361
Valve Spring Guide	N/A	N/A	N/A	3001663	3001663	N/A
Valve Rotator	3081081	3081081	3081081	N/A	N/A	3081081
Valve Spring Retainer	205094	205094	205094	205094	205094	205094
Valve Collet	205091	205091	205091	205091	205091	205091
Valve Stem Seal (4)	N/A	N/A	N/A	N/A	3016692	3016692
Groove Pin	68445	68445	68445	68445	68445	68445
Expansion Plug [7/8 in.]	205401	205401	205401	205401	205401	205401
Expansion Plug [1 5/8 in.]	206224	206224	206224	206224	206224	206224
Expansion Plug [1/4 in.]	203933	203933	203933	203933	203933	203933
Pipe Plug [3/8 in. - 18]	67622	67622	67622	67622	67622	67622
Pipe Plug [1/2 in. - 14]	S-915A	S-915A	S-915A	S-915A	S-915A	S-915A

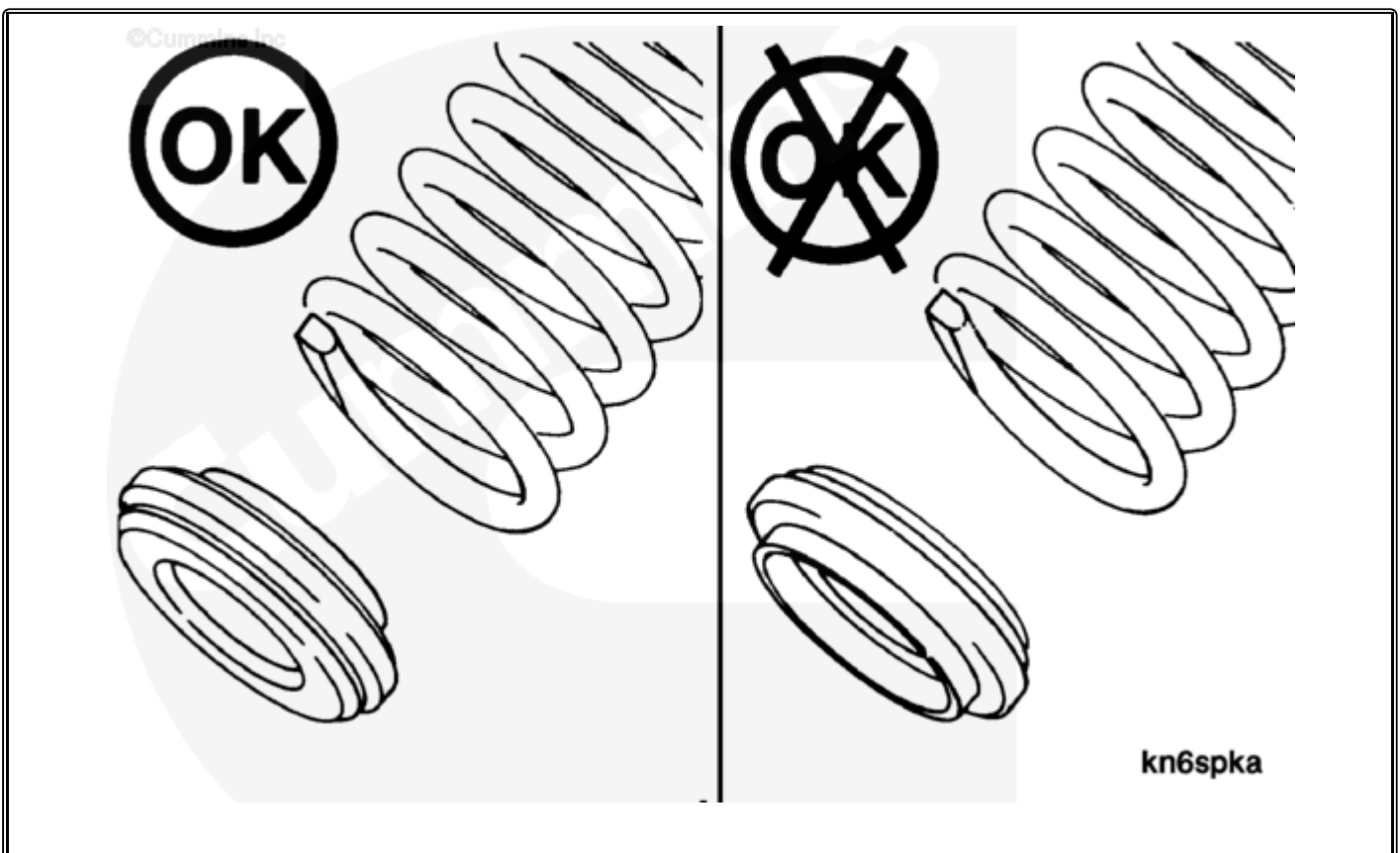
Tribaloy Intake Valve: Intake valve, Part No. 3052820 is available for the K Series engine. This valve is faced with an extremely wear resistant material called Tribaloy. The Tribaloy intake valves are used on specific applications where valve beat-in has been a problem. This intake valve is very effective at reducing the rate of intake valve beat-in when used with rotators and shorter valve springs. These high load constant speed applications include prime and base load generator sets, marine applications, some drilling machines, and hydraulic excavators.

Note : Do not use Tribaloy intake valve, Part No. 3052820, without the Tribaloy intake valve insert, Part No. 3052819 and valve rotator, Part No. 3065936 or 3081081. Also do not use the Tribaloy intake valve in the exhaust position.

The valve head thickness of the Tribaloy intake valve, Part No. 3052820, **must** be measured with the valve head thickness tool, Part No. 3823800. The intake valve head thickness tool, No. 3375940, **must not** be used on the Tribaloy intake valve to prevent grinding the Tribaloy facing too thin.

Valve Spring Rotators: Valve spring rotators were released to decrease valve guide wear. The rotator allows the valve spring to naturally rotate the valve as the valve opens. The rotator prevents the valve from rotating back to the original position. The rotator fits between the cylinder head and the valve spring in place of the lower valve spring guide, Part No. 3001663. Because the rotator adds approximately 9.525 mm [3/8 in] in height to the valve spring assembly, a shorter valve spring **must** be used in conjunction with the valve rotator.

When install a K or KV valve rotator, Part No. 3081081 a rotation check of the rotator in the free state can **not** be done. When turning the rotator by hand, it may turn roughly, appear difficult to turn, or **not** turn at all. Rotators that do not turn freely by hand are **not** "seized". The difficulty in turning the rotator is **not** uncommon. Small dimensional variations in components can make some rotators appear tight, but when the valve spring load is applied, the rotator compresses. This provides more clearance between the collar and the body of the rotator, allowing it to rotate freely.

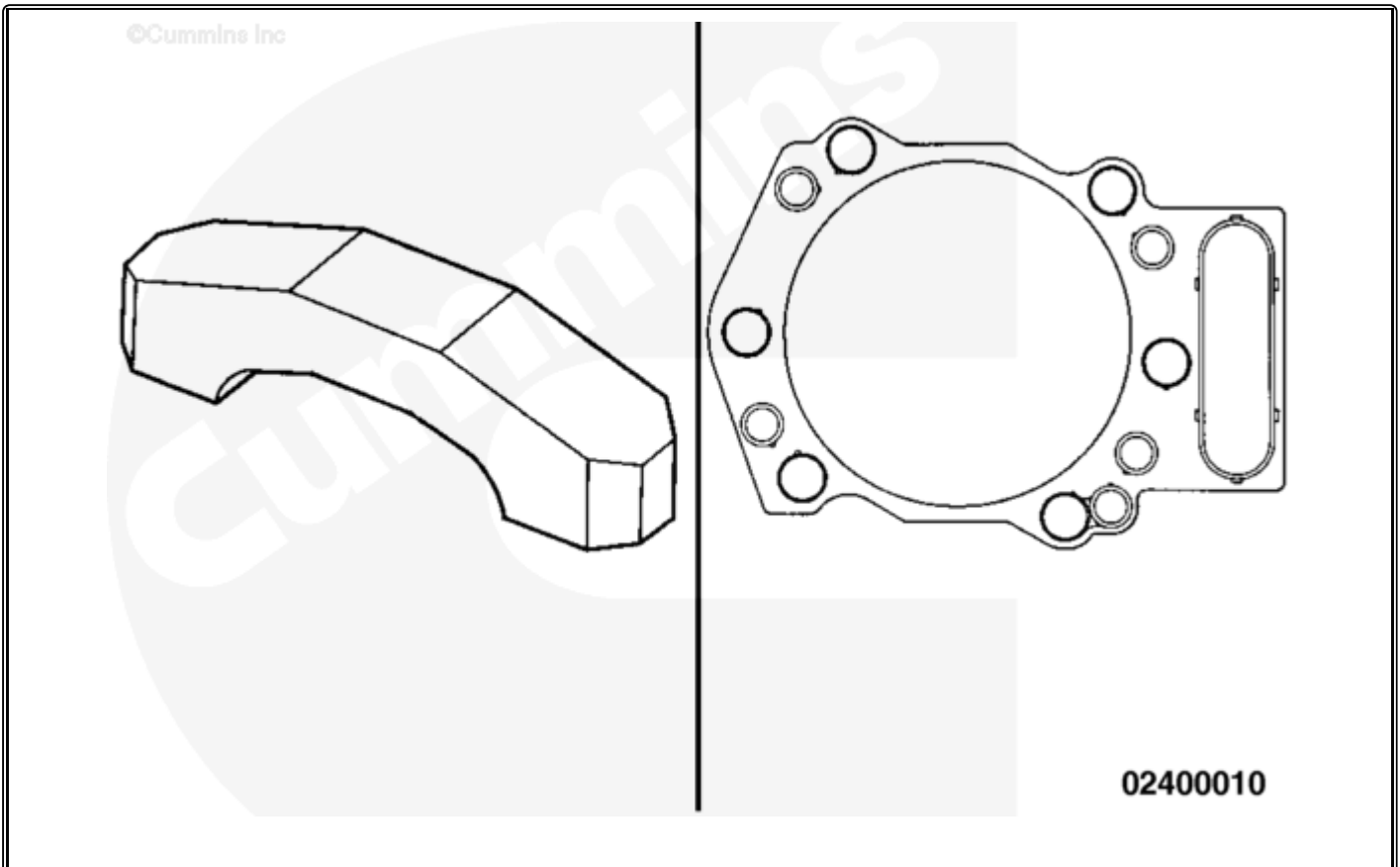


- The rotator **must** be soaked in clean engine oil for at least 5 minutes prior to installation.
- The rotator **must** be installed over the valve guide with the spring pilot flange face-up as shown in this illustration.

Valve Crossheads and Guides: Cummins has released a stemless valve crosshead, Part No. 3086362 for the K Series engines. New cylinder head assemblies, Part No. 3811989, Part No. 3811986, Part No. 3811987, Part No. 3811988, Part No. 3811989, and Part No. 3630361 will **not** have the valve crossheads guide Part no. 205169. The valve crosshead guide hole will still be

machined in the cylinder head casting. Engines using the stemmed crosshead will require installation of the valve crosshead guide Part No. 205169, but the stemless crosshead can be used with or without the crosshead guide installed and can be mixed with guided crossheads on the same cylinder.

Stemless crossheads can be used on all K19, K38 and K50 Series engines. The first K19 engine built with stemless crossheads was ESN first 37158462 on 14Jul1995. The first K38 and K50 engine built with the stemless crossheads was ESN first 33131466 on 07Aug1995. No adjustment is required on the stemless crosshead and it can be installed on the small pocket on either valve. Adjustment of the rocker lever to crosshead lash remains unchanged.



Inconel Exhaust Valve: The Inconel exhaust valve, Part No. 3088389 was released to improve durability of high output and high duty cycle engines. Exhaust valve, Part No. 3088389 supersedes exhaust valve, Part No. 3034827. The Part No. 3088389 exhaust valve has chrome plated stems with the weld moved up into the valve guide. The exhaust valve is very effective at reducing the number of exhaust valve failures in these applications. Uprate engines (more than 101 hp per cylinder) have higher exhaust temperatures and the fatigue strength of the Inconel material increases as the temperature increases. The Inconel exhaust valve, Part No. 3088389 is included in the Premium No. 1, Part No. 3811985, Premium No.2, Part No. 3811986 and the Rail Car, Part No. 3630361 cylinder head assemblies.

Stellite 6 Exhaust Valve: The standard exhaust valve, Part No. 3088391, is faced with Stellite 6 material. Exhaust valve Part No. 3088391 supersedes exhaust valve Part No. 3035110. The Part No. 3088391 exhaust valve has chrome plated stems with the weld moved up into the valve guide. The Stellite 6 material is more heat resistant than the previous Super-Eatonite material. The standard exhaust valve materials can be identified by the cast-in letter code on the valve head. Reference Table VIII, Valve Identification, in this bulletin.

Cylinder Head Castings: The cylinder head castings, Part No. 3009088, 3176619, 3025042, and 3081064, are acceptable for rebuild and reuse. For production only purposes:

- Cylinder head casting, Part No. 3176619 supersedes casting, Part No. 3009088.
- Cylinder head casting, Part No. 3081064 supersedes casting, Part No. 3176619.

When replacing cylinder head castings, Part No. 3009088 or 3176619 with casting, Part No. 3081064, on engines built to CPL-0801, add Field Fix No. 183 to the dataplate.

Cylinder Head castings, Part Nos. 205010, 205015, and 205016, **must** have the injector bore machined to use Premium K injectors.

⚠ CAUTION ⚠

Do NOT machine the injector bores with the cylinder heads installed on the engine. Machining chips will fall into the power cylinder and cause damage to the piston rings and cylinder liner. Any of these castings remanufactured by ReCon® after August 1993 will have the injector bores remachined.

These injector bores **must** be machined to 31.75 to 31.877 mm [1.250 to 1.255 in.] diameter to a depth of 67.564 to 67.818 mm [2.660 to 2.670 in.] from the top deck of the cylinder head.

Cylinder head casting, Part No. 3025042, contains an extra drilling into a coolant passage in the standard cylinder head, Part No. 3176619. This machined casting allows proper venting and coolant flow in horizontally mounted K19 rail car engines. If replacing cylinder head castings, Part No. 3025042, **only** replace with cylinder head casting, Part No. 3081064, on horizontally mounted K19 rail car engines.

Specific generator drive applications require cylinder head casting, Part No. 3050491, Part No. 3068171, or Part No. 3081064 (Reference Table III). Use cylinder head castings, Part No. 3050491, and Part No. 3068171, **only** on those CPL No.s listed in Table III. These castings are **not** to be rebuilt or reused unless on the same engine in the same application from which it was removed. These cylinder head castings can **not** be purchased through ReCon®. If any of the above mentioned CPL No.s require a rebuilt cylinder head, one of the following assembly numbers, which include cylinder head casting, Part No. 3081064 **must** be purchased new.

- Cylinder head assembly, Part No. 3811985 for prime power and base load generator drives.
- Cylinder head assembly, Part No. 3811988 for standby power generator drives only.

TABLE III - Cylinder Head Castings, Part Nos. 3050491 and 3068171, Engine Models

CPL No.	Engine Model	Option No.
CPL-1170	KTTA19-G2	PP4123
CPL-1170	KTTA19-G2	PP4124
CPL-1455	KTA19-G3	PP1455
CPL-1455	KTA19-G3	PP1458
CPL-1251	KTA38-G3	PP6123
CPL-1541	KTA38-G3	PP6188
CPL-1495	KTA38-G4	PP1495

CPL No.	Engine Model	Option No.
CPL-1542	KTA38-G4	PP6189
CPL-1497	KTA38-G5	PP1497
CPL-1543	KTA38-G5	PP6190
CPL-1219	KTA50-G3/G4	PP6121
CPL-1528	K5A50-G3/G4	PP6191
CPL-1254	KTTA50-G2	PP6124C
CPL1544	KTTA50-G2	PP6192

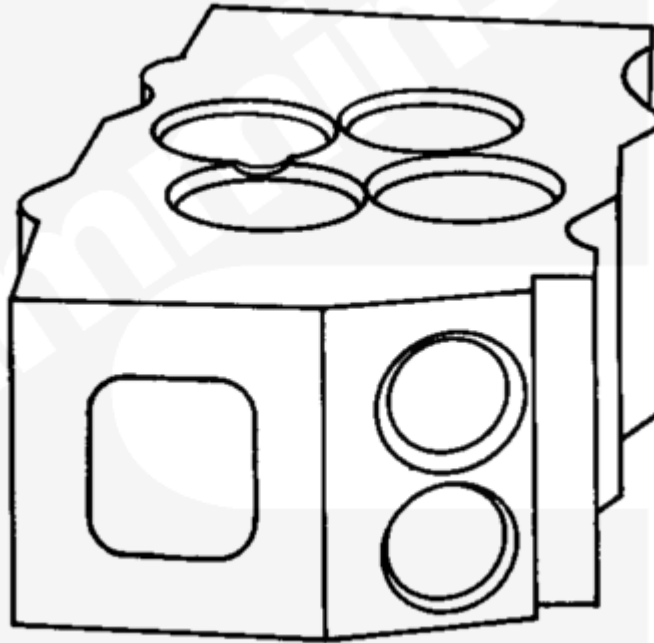
TABLE IV - K Cylinder Head Assemblies Suppression

Head Casting Part No. 3081064 Assembly Numbers	Cylinder Head Casting Assembly Type	Superseded Assembly Part No.
3811985	Premium No. 1	3072438
		3627612
		3627752
		3081070
3811986	Premium No. 2	3074622
		3627753
		3081072
3811987	Improved Standard	3074623
		3081072
3811988	Standard	3068402
		3068404
		3080113
		3081073
3811989	Locomotive (4 Stem Seals)	3015882
		3177529
		3081074
3630361	Horizontal Rail Car	3627997
		3628226
		3081075
		3811990

Cylinder head castings, Part No. 3050491, Part No. 3068171, and Part No. 3081064, can be identified by the cast-in part number and divider wall in the exhaust port. This divider has a 9.525 mm [3/8 in.] horizontal slot on some of the castings with Part No. 3050491 and all of the

castings with Part No. 3068171 and 3081064. The divider wall separates the gases from both exhaust valves. Reference Figure 3 and Figure 4.

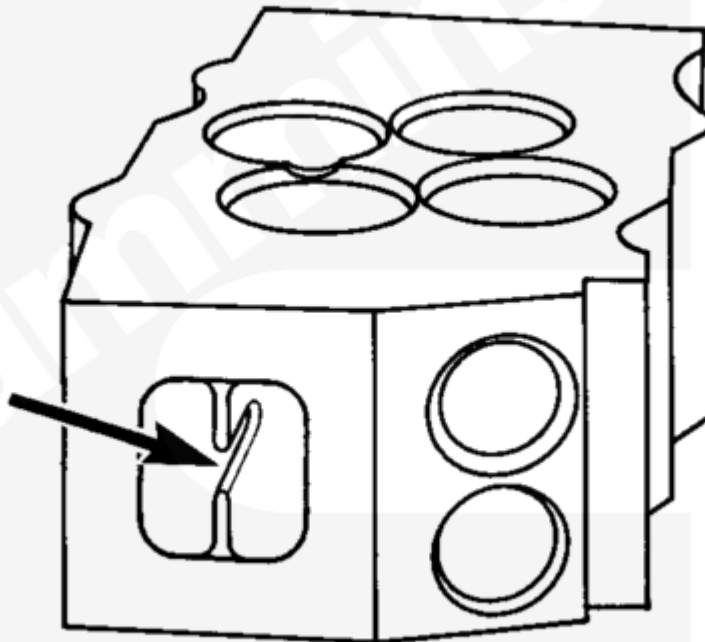
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Figure 3

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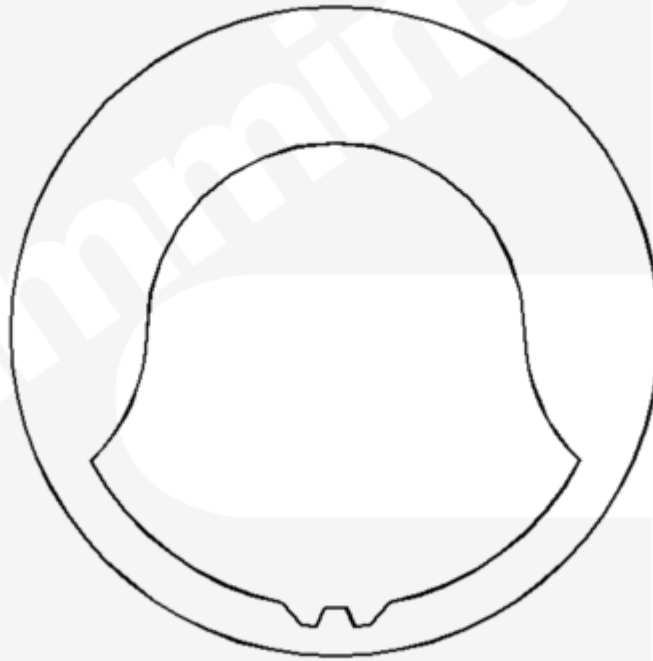
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Figure 4

Anti-Swirl Plates: Anti-swirl plates, also called aid deflectors or swirl plates, are used in some K Series cylinder head assemblies. The plates are installed under the intake valve inserts.

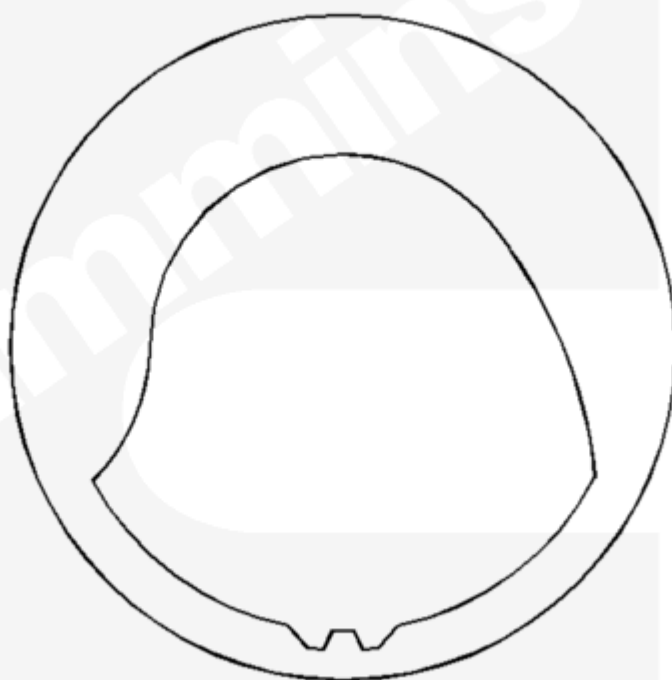
Anti-swirl plates are used to increase performance and reduce emissions by mixing air and fuel. There are two anti-swirl plate types and one spacer plate available for service. In order to meet EPA certifications, certain castings **must** use anti-swirl plates.

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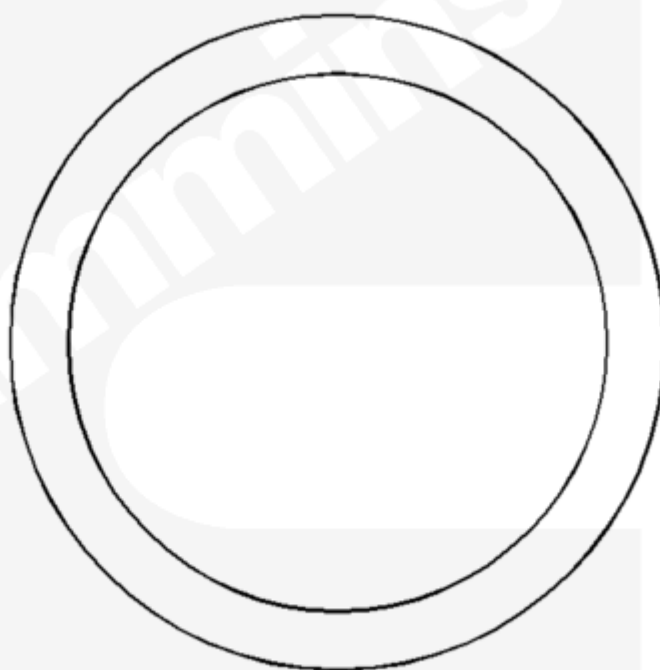
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To meet the 1975 EPA emissions, use anti-swirl plates, Part No. 206262 with cylinder head casting, Part No. 205010, 205015 and 205016.



02600003

Use anti-swirl plate, Part No. 3009109 with cylinder head, casting Part No. 3007207, 3009088, and 3176619.



02600004

Use spacer, Part No. 206813 with cylinder head assembly, Part Nos. AR11368, AT12261 or CPL-127.

The spacer, Part No. 206813 can be used in place of the plate, Part No. 3009109 to meet the 1975 EPA emissions on **only** CPL-159 and CPL-171 when castings, Part No. 3007207, 3009088, or 3176619, are used.

Note : Cylinder head castings, Part Nos 3050491 or 3068171, or the new cylinder head casting does not use any of the above anti-swirl plates or spacer. If the new cylinder head casting is installed on any engine built to CPL-081, stamp Field Fix No. 183 on the anti-swirl dataplate.

The intake valve insert bore depth in cylinder head, Part No. 3050491, 3176619, and 3081064, Figure 8 is **not** as deep as in other cylinder head castings, Figure 9.

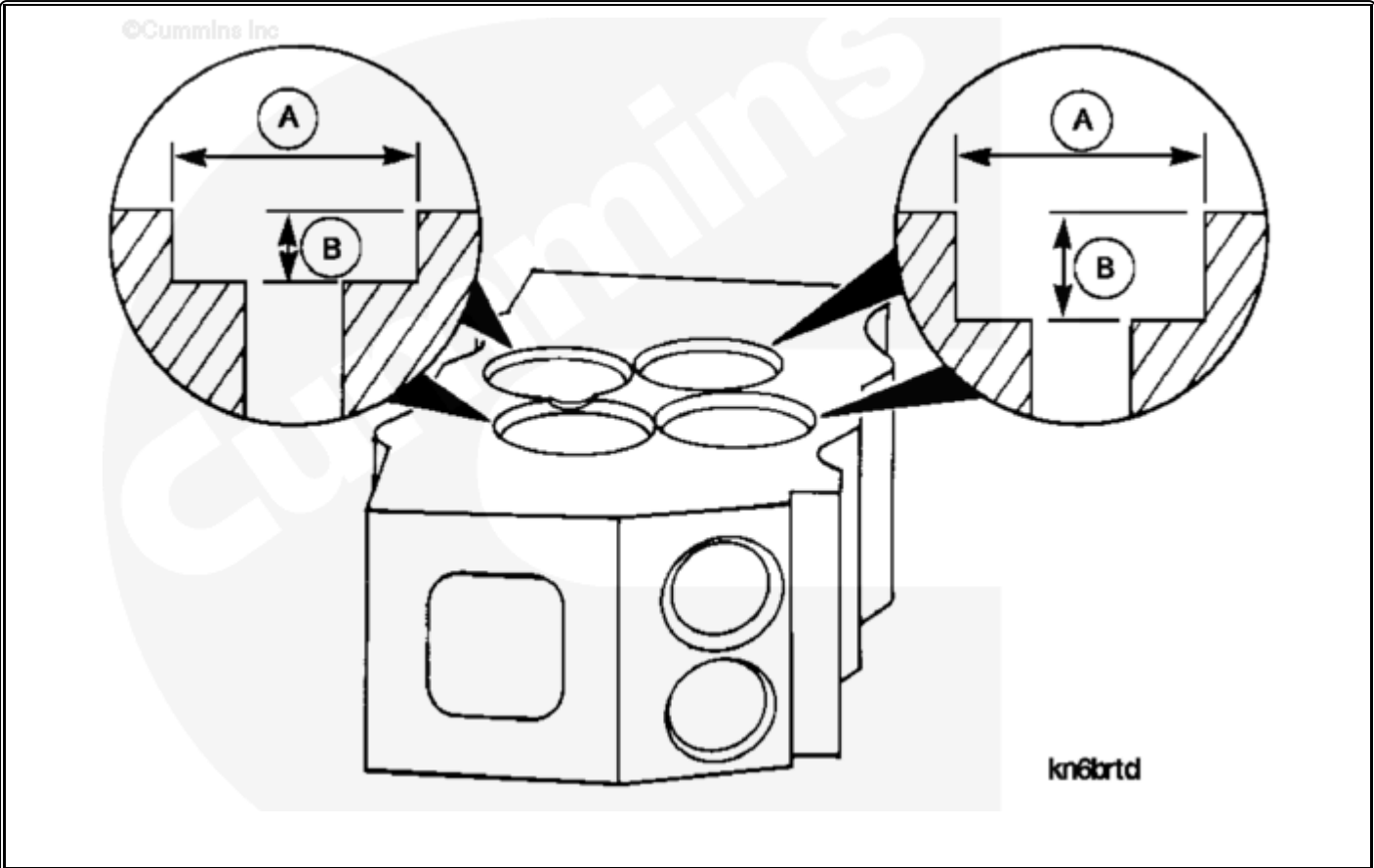


Figure 8. Insert Bore Dimensions with Anti-Swirl Plates

Intake:	Diameter (A)		Depth (B)	
	mm	[in]	mm	[in]
min	60.37	[2.377]	13.00	[0.512]
max	60.40	[2.378]	13.13	[0.517]

Exhaust	Diameter (A)		Depth (B)	
	mm	[in]	mm	[in]
min	60.37	[2.377]	12.50	[0.492]
max	60.40	[2.378]	13.13	[0.517]

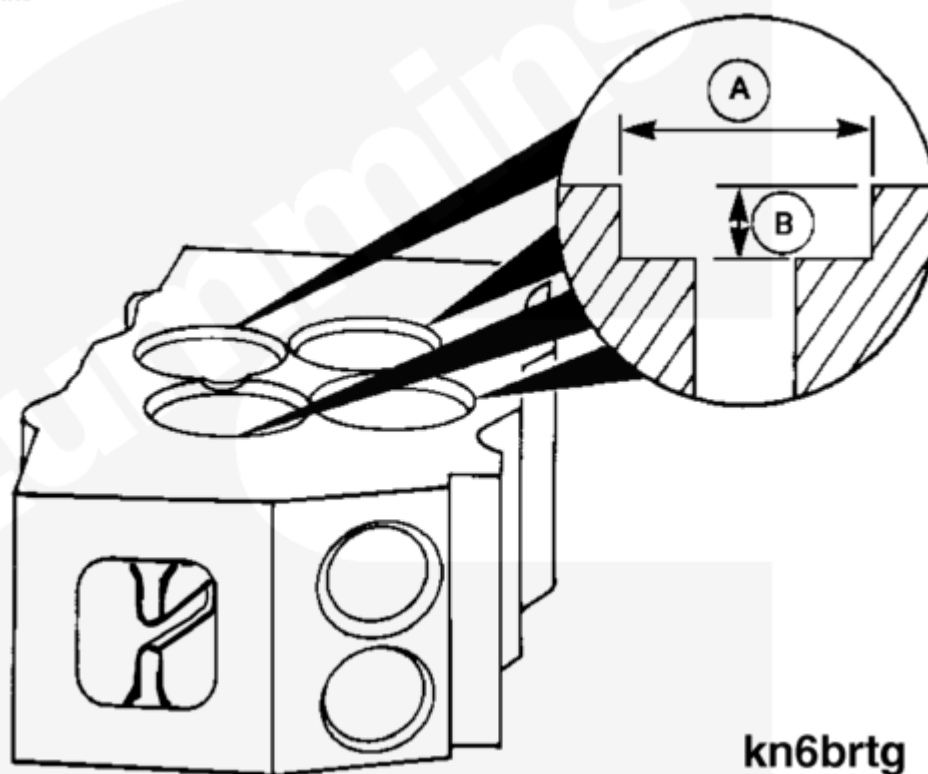


Figure 9. Insert Bore Dimensions without Anti-Swirl Plates

Intake and Exhaust:

	Diameter (A)		Depth (B)	
	mm	[in]	mm	[in]
min	60.37	[2.377]	12.50	[0.492]
max	60.40	[2.378]	12.62	[0.497]

Note : These valve insert bore specifications are for new cylinder heads. The dimensions can be different if the cylinder head has been machined to accept oversized inserts.

Valve Guide Height: Cummins has released new valve guide installation height specifications. There are presently three valve guides released for the K Series engines. The flat top guide is used on applications that require valve guide seals. The taper top guide with an oil hole is used on KT19-R horizontal engines. The guides are to be installed at different heights. Table V supersedes the information found in the K19 Shop Manual, Bulletin No. 3810263, Cylinder Head - Valve Guide - Replace, Procedure 19-4-02-04, page 2-26, and the K38 and K50 Shop Manual, Bulletin No. 3810304, Cylinder Head - Valve Guide - Replace, Procedure 02-04, page 2-18.

Table V - Valve Guide Height

Valve Guide		Installed Height	
		mm	[in]
Flat Top	min	29.21	[1.150]

Table V - Valve Guide Height			
	max	29.72	[1.170]
Taper Top	min	33.66	[1.325]
	max	34.16	[1.345]

Valve Guides: The cast iron, taper top, standard valve guide, Part No. 3201311, has been replaced with valve guide, Part No. 3202210. This guide is made from a porous, oil retaining metal (powdered metal). It does **not** have the internal oil grooves or riflings and has a 45 degree tapered top.

Both the taper top and flat top guides have two oversized guides available. The oversized guides for the K Series engine valve guides are made from cast iron and have the internal oil grooves.

Another current K or KV cast iron valve guide, Part No. 3018818, which is used with a stem seal, protruded into the cylinder head ports more than required. The valve guide length was reduced by 5.59 mm [0.22 in] to 90.93 mm [3.580 in.] to commonized the length with valve guide Part No. 3202210. There is no change to the installed height of the cast iron guide.

Table VI - Taper Top and Flat Top Guides							
45 Degree Taper Top				Flat Top with Seal Groove			
Part No.	Length mm [in]		Outside Diameter mm [in]	Part No.	Length mm [in]		Outside Diameter mm [in]
3202210	90.83 [3.580]		21.48 [0.8458] Standard	3018818	96.52 [3.80]		21.48 [0.8458] Standard
3044972	90.83 [3.580]		21.73 [0.8558] Oversize	3018819	96.52 [3.80]		21.73 [0.8558] Oversize
3044973	90.93 [3.580]		21.86 [0.8608] Oversize	3018820	96.52 [3.80]		21.86 [0.8608] Oversize
Installed Height:				Installed Height:			
	mm		in		mm		in
	33.66	min	[1.325]		29.21	min	[1.150]
	34.16	max	[1.345]		29.72	max	[1.170]

Valve Springs: The reuse specifications for the K Series valve spring have been changed with the release of a new valve spring, Part No. 3630628. This new spring can be identified by a pink stripe painted on it. The previous valve spring, 3629007, which can be identified by a green stripe, white stripe or green spray painted section, **must not** be reused during rebuild. use only Part No. 3630628 valve springs on all K engines during repair or rebuild.

Other valve spring reuse guidelines are as follows:

- A bent or broken spring **must not** be reused.
- A spring with a notch worn in the first coil **must not** be reused. Often, a spring that has these notches will make a CLICK noise. to check for a CLICK, the spring is to be hand compressed until the end of the first coil is completely collapsed. A large notch will cause a CLICK noise to be heard. Check both ends of the spring for this CLICK noise.
- Cummins recommends using the new shrome silicone spring at rebuild, Part No. 3630628.

Table VII, Valve Spring Testing Table, explains the changes in reuse specifications for spring force testing. Reference the K19 Shop Manual, Bulletin No. 3810263, Cylinder Head - Rebuild, Procedure 19-4-02-02, Page 2-11, or the K38 and K50 Shop Manual, Bulletin No. 3810394, Cylinder Head - Rebuild, Procedure 02-02, Page 2-7.

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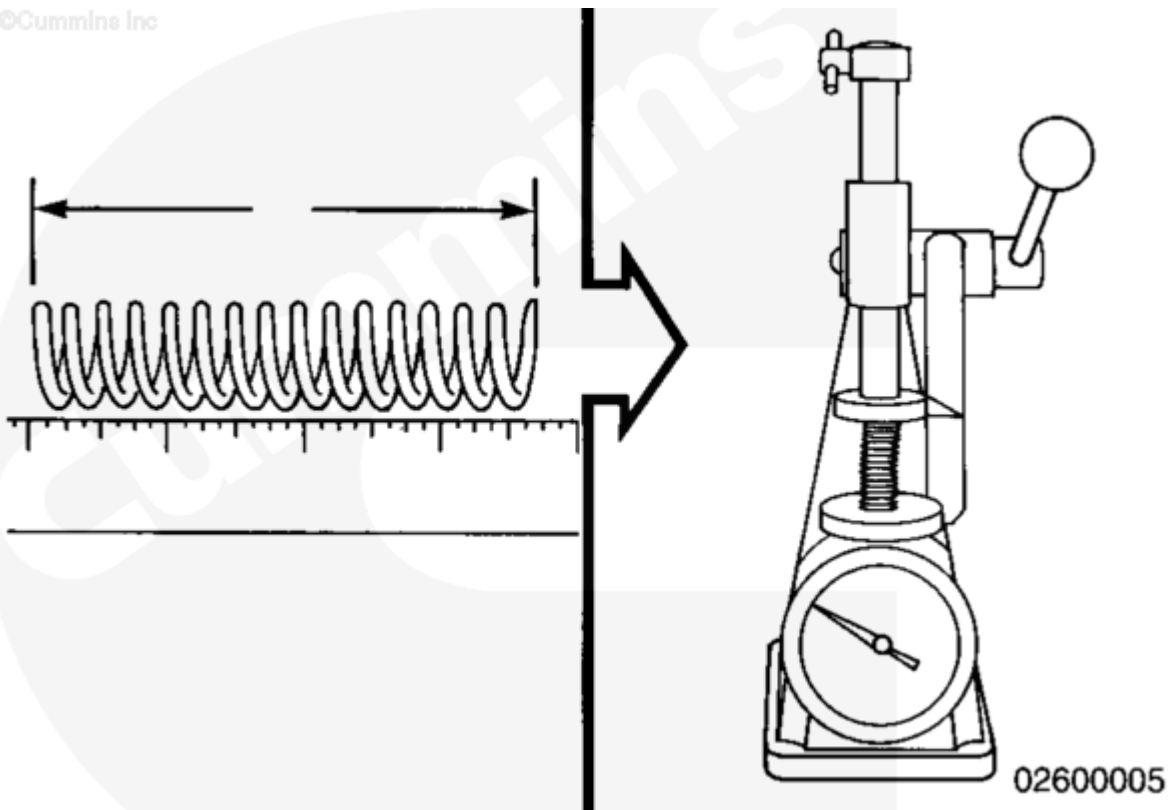
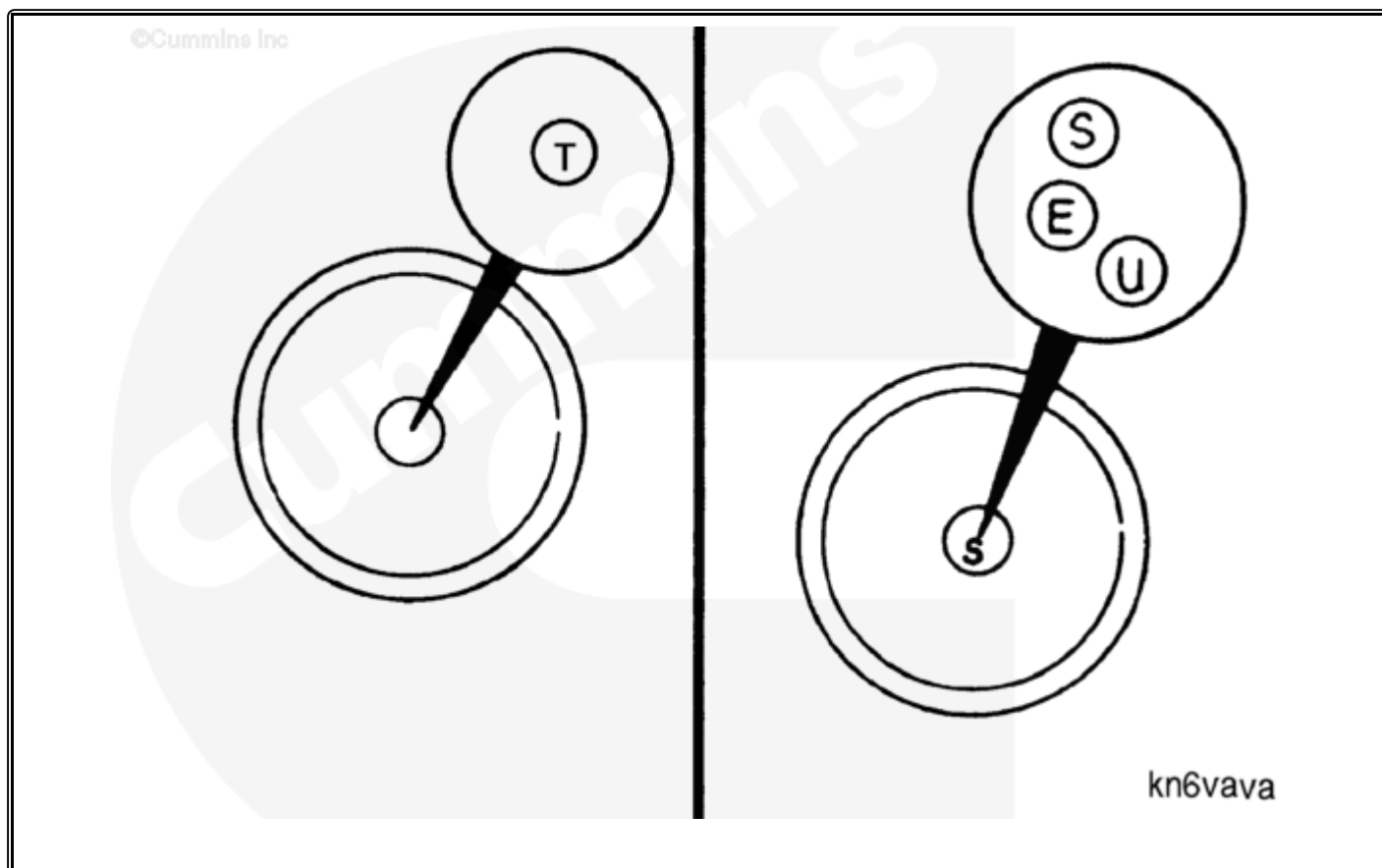


Table VII - Valve Spring Testing Table

Title	Part Nos. 205208, 3629006 Used WITHOUT Rotator	Part Nos. 3065934, 3630628 Used WITH Rotator
Approximate Free Length:		
	85 mm [3.35 in]	67 mm [2.63 in]
Testing Height:		
	50 mm [2.0 in]	38 mm [1.5 in]
Force at Test Height:		
min	1053 N	1134 N
	[235 lbf]	253 [lbf]

Table VII - Valve Spring Testing Table		
max	1237 N	1318 N
	[276 lbf]	294 [lbf]



Valve Identification: There are two methods to identify intake and exhaust valves. Always use the part number for identification when possible. Part numbers are stamped on the valve stem just above the collet groove. A letter code, which is cast into a recession on the combustion face of the valve head, can often be used when the part number is **not** readable.

Table VIII- Valve Identification				
Description	Part No.	Letter Code	Material	Date Chrome Plating Began
Standard Exhaust	3088391	S	Stellite 6	All
Standard Exhaust	3035110	S	Stellite 6	07Sept1995
Standard Exhaust	3035110	E	Super Eatonite 6	None
Premium Exhaust	3088389	U	Inconel	All
Premium Exhaust	3034827	U	Inconel	07Sep1995

Table VIII- Valve Identification				
Premium Intake	3052820	T	Tribaloy*	01May1994
Standard Intake	3088393		Silichrome 1	All
Standard Intake	207241		Silichrome 1	01Jan1996
*Some valves do not have the T letter code or cast recession. In this case, reference the part number.				
Valve Inserts: There are several valve inserts available for service. Four oversized inserts for the exhaust location are available. Two oversized inserts are available for both the standard and Tribaloy intake valves. The location of the part numbers are on the outside diameter of the inserts.				
Table IX - Valve Inserts				
Part No.	Description	Diameter mm [in]	Depth mm [in]	
205093	Exhaust	60.480 mm [2.831 in] Standard	8.052 mm [0.317 in] Standard	
205981	Exhaust	60.734 mm [2.841 in] Oversize	8.052 mm [0.317 in] Standard	
205982	Exhaust	60.988 mm [2.851 in] Oversize	8.179 mm [0.322 in] Oversize	
205983	Exhaust	61.242 mm [2.861 in] Oversize	8.306 mm p0.327 in] Oversize	
205984	Exhaust	61.496 mm [2.871 in] Oversize	8.433 mm [0.317 in] Standard	
3026296	Standard Intake	60.480 mm [2.381 in] Standard	8.052 mm [0.317 in] Standard	
3031098	Standard Intake	60.734 mm [2.391 in] Oversize	8.306 mm [0.327 in] Oversize	
3031099	Standard Intake	60.988 mm [3.401 in] Oversize	8.560 mm [0.337 in] Oversize	
3052819	Tribaloy Intake	60.480 mm [2.381 in] Standard	8.052 mm [0.317 in] Standard	
3052963	Tribaloy Intake	60.734 mm [2.391 in] Oversize	8.306 mm [0.327 in] Oversize	
3052964	Tribaloy Intake	60.988 mm [2.391 in] Oversize	8.560 mm [0.337 in] Oversize	
		min	max	
	Exhaust Seat Width:	1.524 mm [0.060 in]	2.540 mm [0.100 in]	
	Intake Seat Width:	3.048 mm [0.120 in]	3.556 mm [0.140 in]	

Note : The standard exhaust valve, Part No. 3035110 and 3088391 **must** be used with the standard exhaust valve insert, Part No. 205093. The valve rotators can be used with this combination, but only aids the applications where guide wear has been a problem in the past.

Valve Leakage Testing: SPT 91T22-10 was released for a new vacuum leakage testing cup, Part No. 3823852 and seal, Part No. 3823853 for K Series cylinder heads. The new cup and seal was released because the previous design collapsed around the outside diameter of the valve creating an improper No Leakage measurement on the gauge. This new cup and seal will **not** determine if there is a leak between the valve insert and the cylinder head.

Note : Handle the new cup and seal carefully to avoid damage to the new parts.

Before using the tester, test the leakage tester shutoff valve. If the valve is dirty or worn out, it will produce a false leakage measurement on the gauge. Use the following procedure for checking the valve:

- Open the shutoff valve.
- Turn the vacuum pump on.
- Place the cup against a smooth surface.
- Close the shutoff valve.
- Turn the vacuum pump off.
- Wait approximately 10 seconds. The gauge **must not** drop more than 7kPa [2 in Hg].

Cummins ReCon® Cylinder Head Availability: Cummins ReCon® offers four different K engine Series cylinder head assemblies. Listed are the found cylinder head assemblies.

Table X	
Cummins ReCon® Cylinder Head Assemblies	
Part No.	Cylinder Head Description
3021692RX	Standard With Combustion Face Cutouts
3072438RX	Premium No. 1 with all Cylinder Head Castings except 3081064, 3050491, and 3068171
3074622RX	Premium No. 2 with all Cylinder Head Castings except 3081064, 3050491, and 3068171
3074623RX	Improved Standard with all Cylinder Head Castings except 3081064, 3050491, and 3068171
3081070RX	Premium No. 1 with Cylinder Head Casting Part No. 3081064 Only
3081071RX	Premium No. 2 with Cylinder Head Casting, Part No. 3081064 Only

Table X	
Cummins ReCon® Cylinder Head Assemblies	
Part No.	Cylinder Head Description
3081072RX	Improved Standard with Cylinder Head Casting, Part No. 3081064 Only
3604153RX	No Combustion Face Cutouts, Automotive Only

Cummins ReCon® does **not** offer the Improved Standard Cylinder Head Assembly, Part No. 3081072. Industrial (construction, power unit, locomotive applications requiring this cylinder head **must** upgrade to the Premium No. 2 cylinder head assembly, Part No. 3081071RX, if a ReCon® cylinder head is desired.

Cylinder Head Interchange: Do **not** downgrade cylinder head assemblies such as replacing Premium No. 1 cylinder head assembly, Part No. 3081070, with Premium No. 2 cylinder head assembly, Part No. 3081071. Upgrading of cylinder head assemblies is allowable. Reference able IV for cylinder head assembly supersession. A cylinder head assembly containing the new cylinder head casting, Part No. 3081064 can be mixed on the same engine with superseded cylinder head assemblies containing an older part number casting. Before replacing a cylinder head assembly that has a new casting with an assembly that has the old casting, consult the CPL manual. If the new casting part number is the **only** part number given in the CYLINDER HEAD field of the CPL information, use **only** a cylinder head assembly with a new casting as replacement. On some later CPLs, if the new casting is replaced with an old casting, the engine can suffer some performance (fuel consumption) degradation and exhibit more black exhaust smoke.

The cylinder head assembly, Part No. 3604153RX is to be used **only** on K19 automotive engines. It is marketed specifically to reduce white smoke complains for automotive K19 engines operating in cold ambient temperatures. Cylinder head assemblies 3074622RX and 3081071RX feature Inconel exhaust valves with chrome stems and standard intake valves. Cylinder head assemblies 3072438RX and 3081070RX features Tribaloy intake valves and Inconel exhaust valves with chrome stems. Cylinder head assemblies 3081072RX and 3074623RX were released 01Jun1996. They feature Stellite valve and chrome valve stems. All of the heads listed in Table X will have chrome valve stems implemented by 01Aug1996, except for older model heads 3021692RX and 3604153RX.

These are product improvements **not** subject to campaign.

Document History

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